A Virtual Bank Application for Kids

-developing the software using Agile Methods

1. General information

In the next few weeks, your team will be required to develop a Virtual Bank Application for Kids using Agile methods. Your team should aim to deliver a simple first release of the software product that can be extended in further iterations. Agile methods should be applied in all activities, from requirements through analysis/design, implementation, and testing. Iterations should be planned, and outcomes should be submitted.

There are no restrictions on what the Virtual Bank Application for Kids should include, and the given specification contains only high-level abstract requirements. It should be noted that determining the software requirements is one of the most important and complex phases in any development project.

You should apply requirement-finding techniques and Agile methods to identify the actual requirements at an appropriate level.

Most importantly, you need to prioritize the features that are implemented in accordance with both ease of implementation and meeting requirements. As in real software, you should define the project scope properly. Keep your design SIMPLE. Bear in mind that there is no absolute right answer – your solution may be perfectly appropriate.

2. **First submission:** Product backlog and Prototype, 29th March, 2026
   Final submission: Report and Software: 10th May, 2026
   Demonstration: 11th – 16th May, 2026
   Marks returned: Approximately 2-3 weeks after the final submission.

Specification of the project

3.1 Basic requirements

Your team will develop a fun and easy to use "Virtual Bank Application for Kids". The objective of this application is to educate children about the value of money and the concept of a bank, encouraging them to engage in tasks to earn pocket money, save for goals, and spend responsibly. There are no restrictions on the functions the application should include, as one of the most important tasks of this project is to identify the actual requirements. Here are some suggested functions to assist you in getting started.

• *Account creation:* create virtual bank accounts, including current accounts

and saving accounts.
• *Balance tracking:* display the current balance.
• *Deposit:* enable kids to deposit virtual money they earned from doing tasks.
• *Withdrawal:* enable kids to withdraw virtual money.
• *Task setting:* parents can set tasks or activities (e.g. house chore, exercises) to give kids opportunities to earn money.
• *Transactions:* can check the transaction history.
• *Savings goals:* enable kids to set savings goals and to track progress towards those goals.
• Any other function(s) that is useful.

A full prototype of the application should be produced. It is not required to implement the full working code, however your team should implement core functions of your choice.

3.2 Other requirements

• The software must be developed using Java as a stand-alone application running on computers. A simple graphic user interface (GUI) should be used. The recent Java Edition should be used (for ex. JavaFX). **Do NOT BUILD a Web-based application or Phone App.**
• The application should be used without an Internet connection.

- Only API can be used for connection between a child and parents (but not mandatory)

• All input and output data should be in simple text file format. You may use plain text (txt), CSV, JSON, or XML. **Do NOT use a database.**
• Basic restrictions and error checking must be considered.
• Your design must be flexible and extensible to adapt to future changes, e.g. modify existing features and add new features. When doing so, you should be able to reuse the existing components and make the least impact on the existing code.

Your tasks are to define detailed requirements, design, develop and test the above described software using Agile methods. Feel free to design the software as long as it satisfies the basic requirements, define the SCOPE properly.

3. Agile project management

Each group has 4-5 students. You are the Agile team working together to complete the project. All students in a group must work on ALL aspects of the project, to obtain full software engineering skills. You should use the techniques you have learnt in the lectures to manage the project, e.g., Scrum, daily stand up meetings, working around a table, scrum master and decision making, etc. You are also encouraged to use other efficient ways of communication to coordinate the group activities.

Suggested Timeline:

- 16th -20th Feb, 2026

Activities: meet group members, appoint a group leader and discuss the project handout.

- 23rd Feb - 13th March, 2026

Activities: gather actual requirements, story writing workshop.
Outcomes: product backlog.

- 16th - 27th March, 2026

Activities: create prototype and get user feedback.
Outcomes: product prototype.

- 1st - 10th April, 2026

Activities: Iteration 1.
Outcomes: Working Software v1.

- 13th - 17th April, 2026

Activities: Iteration 2.
Outcomes: Working Software v2.

- 20th – 24th April, 2026

Activities: Iteration 3.
Outcomes: Working Software v3.

- 27th April- 8th May, 2026

Software and report final delivery preparation.

Submissions on email: sherzod.haydarov@polito.uz

For all the submissions, only the group leader should submit the files on behalf of the whole group.

The first submission includes product backlog and prototype. 29th March, 2026

The final submission includes a report and software. 10th May, 2026

4.1 The product backlog, an excel file (refer to the template on LMS). Filename: Productbacklog_groupXXX.xlsx, where XXX is your group number. It should contain all user stories with acceptance criteria, priority, estimation and iteration plan.

4.2 The prototype, a PDF file. Filename: Prototype_groupXXX.pdf, where XXX is your group number. It should contain full prototype. Only low-fidelity or medium fidelity prototype is needed.

Find more about what is a prototype through this link:
https://tirapid.com/what-is-prototype-in-software-engineering/

4.3 The report, a PDF file. Filename: Report_groupXXX.pdf, where XXX is your group number. The report template provided must be used. It should contain the sections of Group report (maximum 25 pages including tables, charts, figures and diagrams you may have) and individual statements (no more than 300 words each). More details can be found in the template.

4.4 The software, a ZIP file. Filename: Software_groupXXX.zip, where XXX is your group number.

It should contain the following parts:

a) Java code. All core functions should be implemented. Code should be well documented.

b) A set of test programs using Junit as an example of using TDD.

c) JavaDocs.

d) A user manual with some key screenshots of the application.

e) A readme file to instruct how to set up or configure and run your software.

4. **Demonstration**

   Informal demonstration (not assessed):

   There are two informal demonstrations, and the main purpose is to gather feedback. You should arrange a 30-minute session with your group's leader.

   a) Demo 1: during week 16th - 27th March, 2026

   b) Demo 2: during week 13th - 17th April, 2026

   Formal demonstration (assessed): 11th – 16th May, 2026

Your team is required to formally demonstrate the final software product (assessed).

5. ALL group members MUST attend the demonstration session. You should showcase the functionality of your software product as if you were presenting it for sale.

Demonstrate the operation of the core functions and, if possible, exhibit error handling capabilities.
Additionally, be prepared to answer a few questions.